

STAT 516: Basic Probability and its Applications

Lecture 1: Introduction to Probability. Sample spaces, events, probability axioms

Random Experiments and a Sample Space

- ▶ A **random experiment** can have several possible **outcomes**
- ▶ The **sample space** of a random experiment $\mathcal{S}$ is the set of all its possible outcomes.
- ▶ Example: tossing the coin once we have the space $\mathcal{S} = \{H, T\}$
- ▶ Example: tossing the coin twice and recording the outcomes produces $\mathcal{S} = \{HH, HT, TH, TT\}$
- ▶ Example: roll a die with 6 faces and record the outcome. Then, $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$.

The idea of probability

- ▶ For us, **probability** is the idealized quantity the **relative frequency** approaches as the number of experiments grows
- ▶ Other approaches are possible...e.g. treating the probability as as personal (subjective) statement of beliefs
- ▶ The theory does not change

Basic Set theory I

- ▶ A **set** is a collection of objects
- ▶ Example: if $\mathbb{R}$ is a set of all real numbers , then 0.5 is an **element** of $\mathbb{R}$
- ▶ Notation: $0.5 \in \mathbb{R}$
- ▶ If $C = \{x : 0 \leq x \leq 1\}$, we say that $0.5 \in C$ as well
- ▶ A set C is **countable** if it can be **enumerated**

Examples

- ▶ Neither $\mathbb{C}$ nor $\mathbb{R}$ are countable
- ▶ The set $\{1, 2, 3, 4, 5\}$ is countable
- ▶ The set of all integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$ is also countable

Basic Set theory II

- ▶ A is a subset of B if for every $x \in A$ we have $x \in B$
- ▶ Notation: $A \subset B$
- ▶ If $A \subset B$ and $B \subset A$, then $A = B$
- ▶ A set A is an **empty set** if it has no elements.
- ▶ Notation: $A = \emptyset$

Some Relations from Set Theory

- ▶ The **union** of two events A and B is the event consisting of all outcomes that are *either* in A or in B or in *both* events. Notation: $A \cup B$. Reading: A or B
- ▶ The **intersection** of two events A and B is the event consisting of all outcomes that are in *both* A and B . Notation $A \cap B$. It is read " A and B ".
- ▶ The **complement** of an event A , denoted by A' , is the set of all outcomes in $\mathcal{S}$ that are *not* in A .

Some Relations from the Set Theory

- ▶ The set $A \triangle B$ consists of all outcomes that are either in A or in B but *not in both*. Reading: a **symmetric difference** of A and B
- ▶ The **difference** between two sets A and B or a **relative complement of B in A** $A \setminus B = A \cap B'$ consists of all outcomes in A that are *not* in B
- ▶ The notation $A - B$ is also used but is less preferred because it can also mean the set of all $\{a - b | a \in A, b \in B\}$
- ▶ As an example, $A \cup B \cup C$ means at least one of A , B or C occurred
- ▶ $A \cap B \cap C$ means that all of A , B and C occurred
- ▶ $A \cap B' \cap C'$ means that A occurred but not B or C

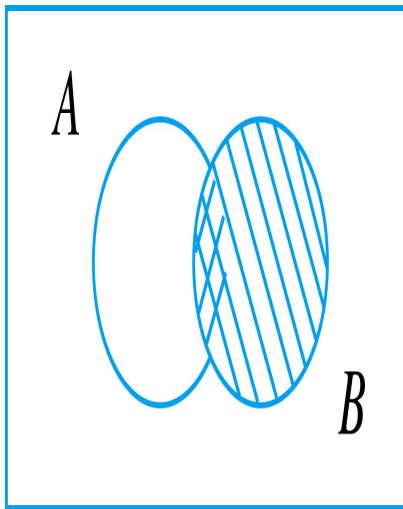
Examples

- ▶ If $C_1 = \emptyset$, $C_1 \cup C_2 = C_2$ for any C_2
- ▶ For any C $C \cup C = C$
- ▶ $C_1 = \{1, 2, 3\}$ and $C_2 = \{2, 3, 4\}$
- ▶ $C_1 \cup C_2 = \{1, 2, 3, 4\}$ and $C_1 \cap C_2 = \{2, 3\}$
- ▶ Consider rolling a die with the sample space $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$.
- ▶ $A' = \{4, 5, 6\}$.

Some Additional Set Theory Relations

- ▶ For a sample space $\mathcal{S}$, an element $x \in \cup_k A_k$ if and only if there exists k_0 such that $x \in A_{k_0}$.
- ▶ $x \in \cap_k A_k$ if and only $x \in A_k$ for all k
- ▶ Example: if $A_k = \{0, 1, 2, \dots, k\}$ then $\cup_k A_k = \{0, 1, 2, 3, \dots\}$ and $\cap_k A_k = \{0, 1\}$
- ▶ Example: let $C_k = \left\{x : \frac{1}{k+1} \leq x \leq 1\right\}$, $k = 1, 2, 3, \dots$. Then, $\cup_{k=1}^{\infty} C_k = \{x : 0 \leq x \leq 1\}$
- ▶ Example: let $C_k = \left\{x : 0 < x < \frac{1}{k}\right\}$ for $k = 1, 2, 3, \dots$. Then, $\cap_{k=1}^{\infty} C_k = \emptyset$.

Venn Diagrams



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Some operation laws on sets

► **Distributive laws:**

$$B \cap (\cup_k A_k) = \cup_k (B \cap A_k)$$

and

$$B \cup (\cap_k A_k) = \cap_k (B \cup A_k)$$

► **De Morgan's laws:** for any two sets A_1 and A_2

$$[A_1 \cap A_2]' = A_1' \cup A_2'$$

and

$$[A_1 \cup A_2]' = A_1' \cap A_2'$$

Set functions

- ▶ **Set functions** are important in probability!
- ▶ Example: C is a set in a two-dimensional space and $Q(C)$ is its area if C is finite
- ▶ For a circle $C = \{(x, y) : x^2 + y^2 \leq 1\}$ $Q(C) = \pi$
- ▶ For a set $C = \{(0, 0), (1, 1)\}$ $Q(C) = 0$
- ▶ For a rectangular triangle
 $C = \{(x, y) : 0 \leq x, 0 \leq y, x + y \leq 1\}$ $Q(C) = \frac{1}{2}$

Summation for set functions

- ▶ Let

$$f(x) = \begin{cases} \left(\frac{1}{2}\right)^x & \text{if } x = 1, 2, 3, \dots \\ 0 & \text{elsewhere} \end{cases}$$

- ▶ For $C = \{0 \leq x \leq 3\}$,

$$Q(C) = \sum_C f(x) = \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 = \frac{7}{8}$$

Integration for set functions

- ▶ Let $Q(C) = \int_C \exp(-x) dx$
- ▶ For $C = \{x : 0 \leq x < \infty\}$

$$Q(C) = \int_0^{\infty} \exp(-x) dx = 1$$